

		left must be provided by the ground. That is only possible if the ground is rough so as to provide a frictional force on the ladder.
8	B	$U_w + U_m = W$ $\rho_w V_w g + \rho_m V_m g = \rho_n V_n g = \rho_n (V_w + V_m) g$ $1000 V_w + 13600 V_m = 7900 (V_w + V_m)$ $5700 V_m = 6900 V_w$ $V_w / V_m = 57/69 = 0.826$
9	B	The only forces acting on the mass are the tension and the weight, which combines to a resultant centripetal force.